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Unmasking Cyber Threats and Fortifying your Microsoft Environment

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We live in the most complex threat landscape in history



775M

Email messages containing malware



Password attacks per second



275%

Increase in human-operated ransomware attacks from 2022 to 2024



<1B

+408% YoY

Victims of data breaches keep increasing

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Technical Challenges



Technical Debt



Shadow IT



Complexity



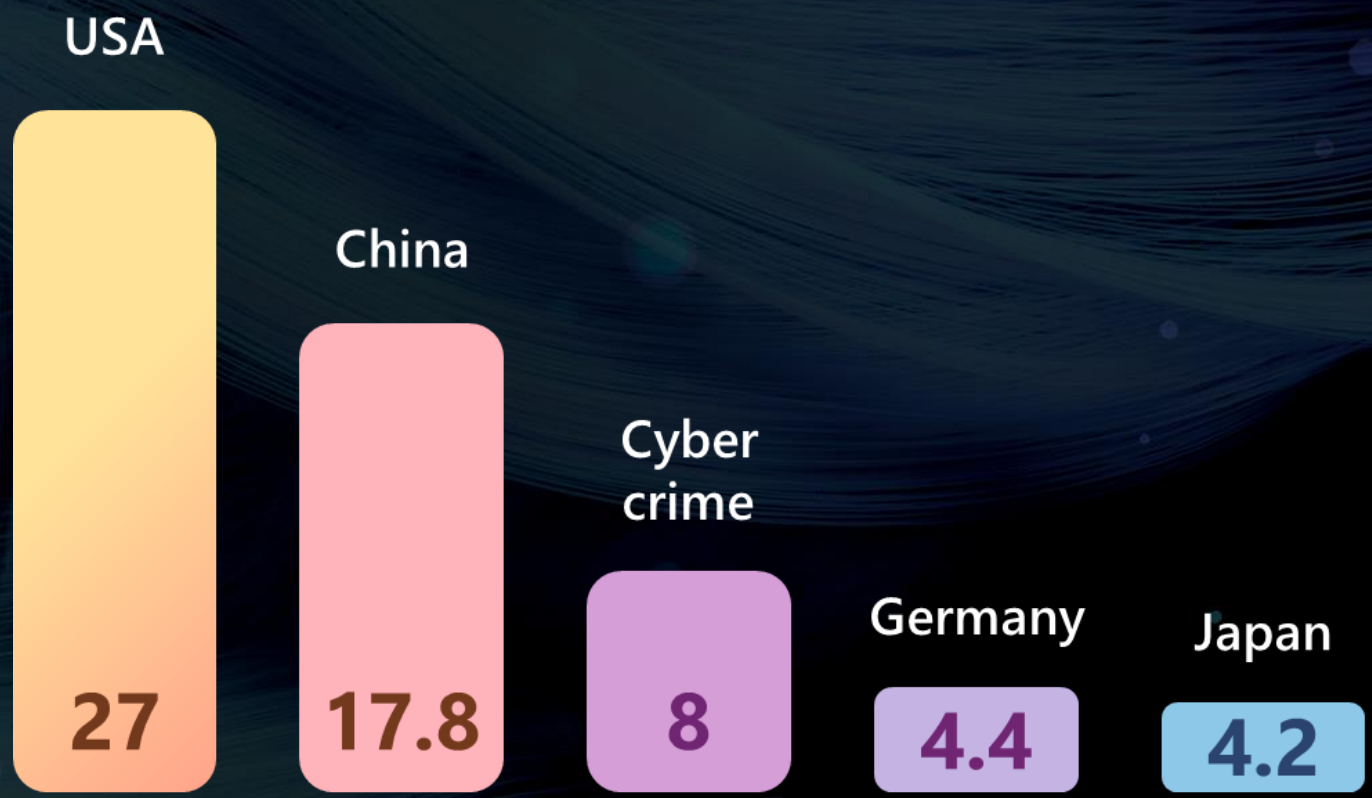
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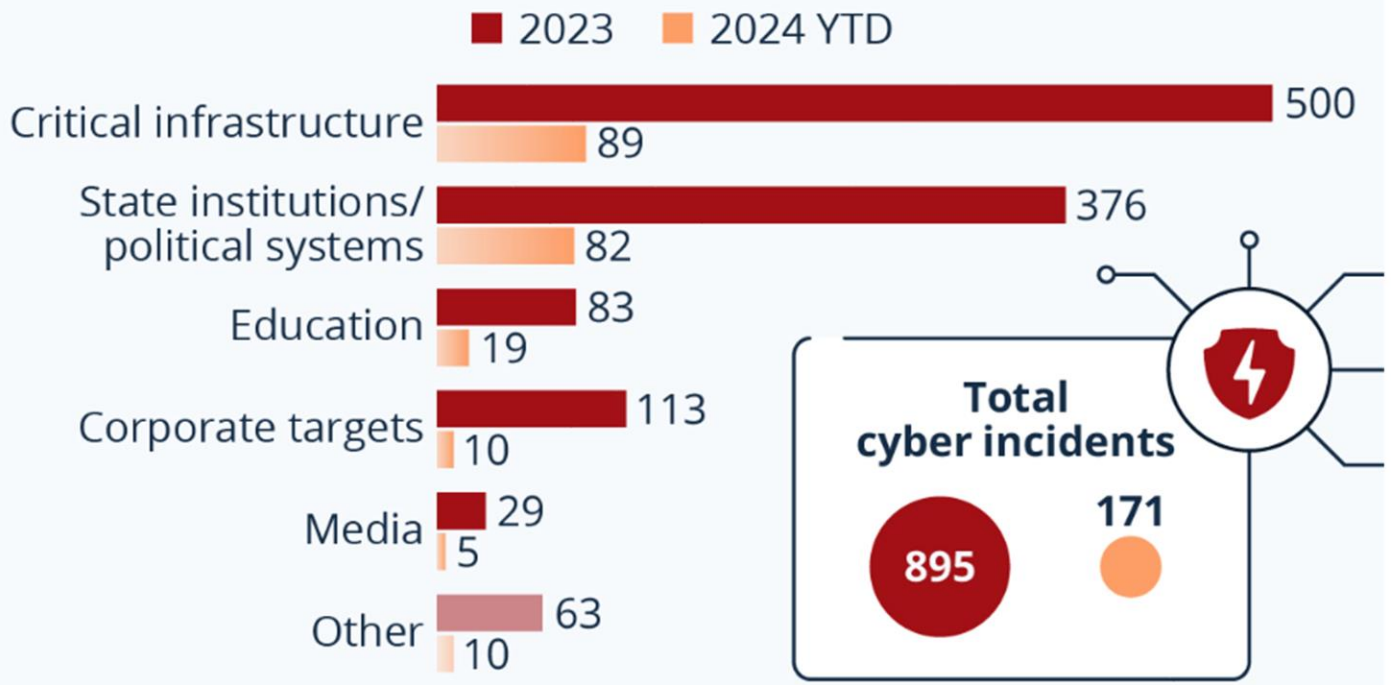
Cybercrime is the 3rd largest GDP

Annual GDP, in trillions of USD



Cybercrime: Critical Infrastructure Is Top Target

Number of worldwide political cyber attacks aimed at the following sectors, by reported period*



As of Mar. 26, 2024

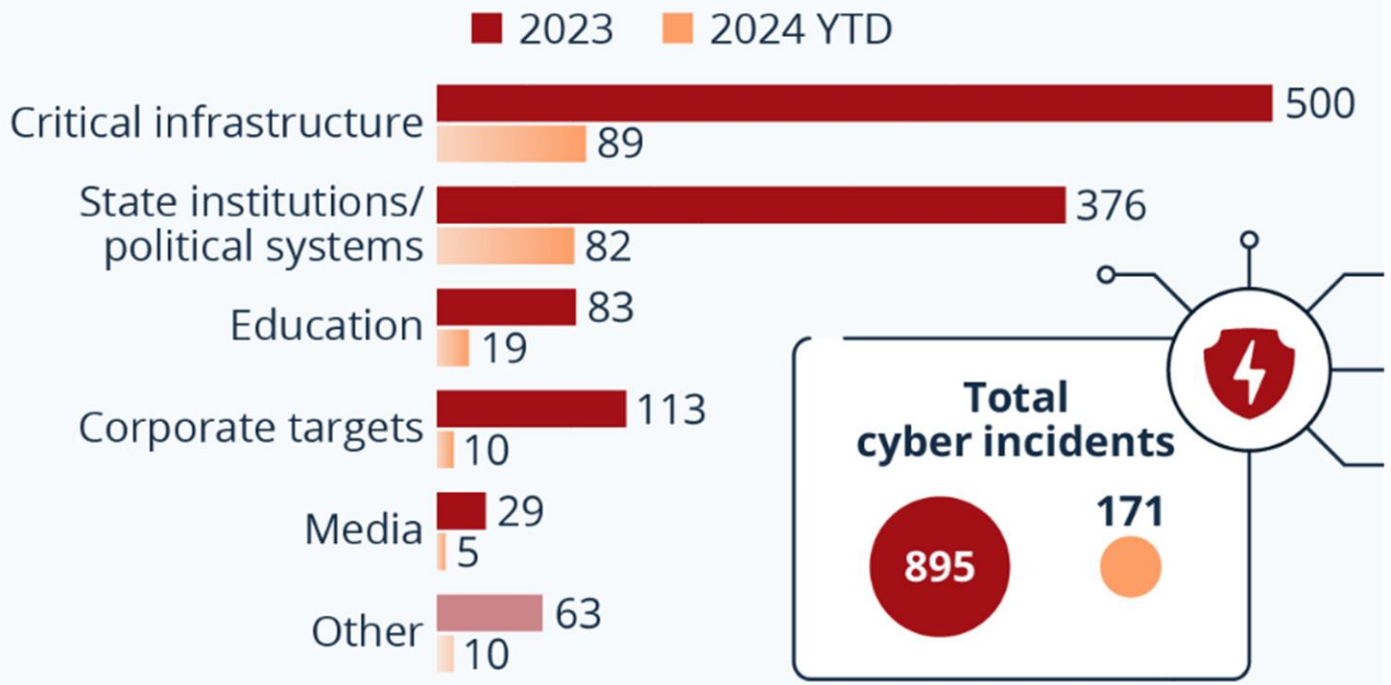
* One incident may target more than one sector. Initial access by malevolent actors may not always coincide with report period.

Source: European Repository of Cyber Incidents



Cybercrime: Critical Infrastructure Is Top Target

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Advance Persistent Threats (APT)

APT38



State-Sponsored

Famous Chollima

NoName057



Hacktivism

Bounty Jackal

Akira



Criminal

Punk Spider

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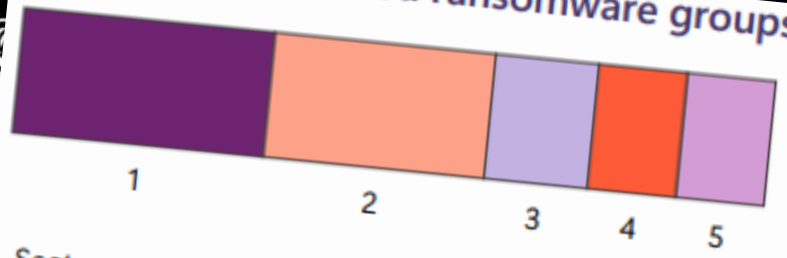
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Weapons Unmasked

Ransomware



Top human-operated ransomware groups



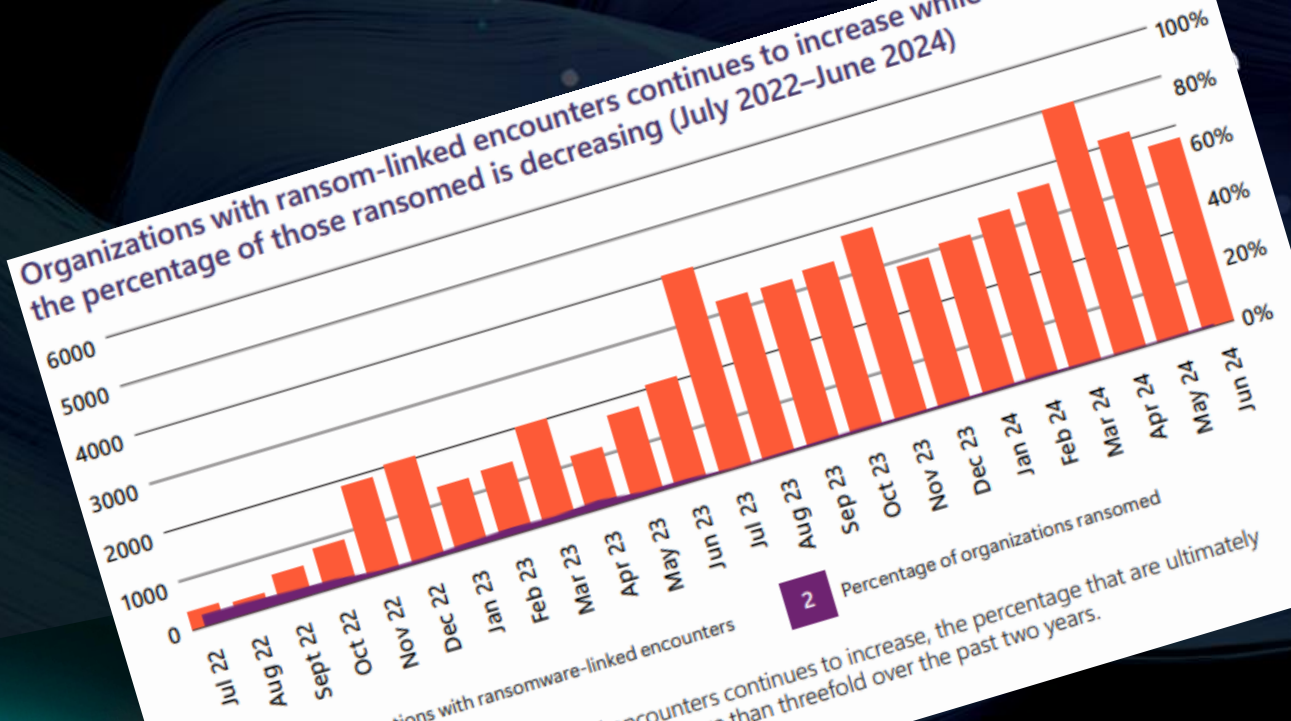
	Sector	Percentage
1	Akira	17%
2	Lockbit	15%
3	Play	7%
4	Blackcat	6%
5	Basta	6%

The top five ransomware families accounted for 51% of attacks. These families continue to use longstanding techniques, showing their effectiveness even against rising cybersecurity awareness globally.

Source: Microsoft Defender for Endpoint



Organizations with ransom-linked encounters continues to increase while the percentage of those ransomed is decreasing (July 2022–June 2024)



1 Number of organizations with ransomware-linked encounters

2 Percentage of organizations ransomed

Although organizations with ransom-linked encounters continues to increase, the percentage that are ultimately ransomed (reaching encryption stage) decreased more than threefold over the past two years.

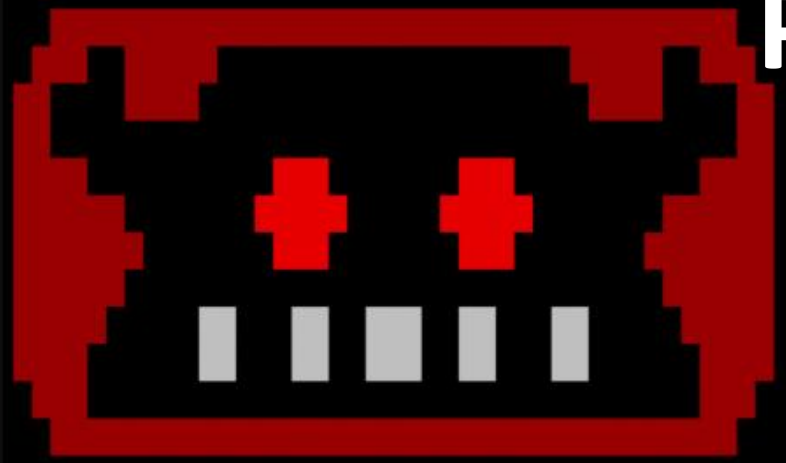
Source: Microsoft Defender for Endpoint


Tampering
with
Security Products

Phishing



Phishing



```
[[                  :~]# evilginx -debug

 _____ _____ _____ __    _____ _____ _____ _____
|   __|  |  |     |  |  |   __|     |   | |  |  |
|   __|  |  |-   -|  |__|  |  |-   -| | | |\   /|
|_____|\___/|_____|_____|_____|_____|_|___|_| |_|

                no nginx - pure evil

        by Kuba Gretzky (@mrgretzky)     version 2.3.3

[16:18:50] [inf] debug output enabled
[16:18:50] [inf] loading phishlets from: /usr/share/evilginx/phishlets/
[16:18:50] [inf] loading configuration from: /root/.evilginx
[16:18:50] [war] server domain not set! type: config domain <domain>
[16:18:50] [war] server ip not set! type: config ip <ip_address>

+------------------+------------------+----------+------------+----------+
|    phishlet      |      author      |  active  |   status   | hostname |
+------------------+------------------+----------+------------+----------+
| amazon           | @customsync      | disabled | available  |          |
| instagram        | @prrrrinncee     | disabled | available  |          |
| outlook          | @mrgretzky       | disabled | available  |          |
| reddit           | @customsync      | disabled | available  |          |
| twitter-mobile   | @white_fi        | disabled | available  |          |
| facebook         | @mrgretzky       | disabled | available  |          |
| o365             | @jamescullum     | disabled | available  |          |
| okta             | @mikesiegel      | disabled | available  |          |
| protonmail       | @jamescullum     | disabled | available  |          |
| twitter          | @white_fi        | disabled | available  |          |
| linkedin         | @mrgretzky       | disabled | available  |          |
| wordpress.org    | @meitar          | disabled | available  |          |
| citrix           | @424f424f        | disabled | available  |          |
| github           | @audibleblink    | disabled | available  |          |
+------------------+------------------+----------+------------+----------+
```

Identity attacks in perspective

Password-based attacks continue to dominate, but can be thwarted by using strong authentication methods.





Less than 1% combined

MFA attacks

- SIM swapping
- MFA fatigue
- AitM

End-run MFA protection by intercepting security codes using stolen phone numbers, barraging users with MFA notifications until they approve, and capturing first and second factor credentials using fake replicas of legitimate websites.

Post-authentication attacks

- Token theft
- Consent phishing

Infiltrate a user's account after they authenticate by stealing a legitimate token created on their device and moving it to a device under the attacker's control, by searching source code repositories for Open Authorization (OAuth) tokens and other non-human credentials, or by tricking the authenticated user into granting permissions to malicious apps.

Infrastructure compromise

Often silently executed by professional groups or nation-state-backed threat actors with sophisticated operations, making them very hard to detect. Threat actors may compromise an on-premises federation server and copy its private signing key to forge tokens, compromise a privileged cloud user and add new federation contracts, or compromise a non-human workload identity and create new credentials with elevated privileges.

Source: Microsoft Threat Intelligence

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Threat Frameworks

ATT&CKcon 6.0 returns October 14-15, 2025 in McLean, VA. More details about tickets and our CFP can be found here

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MITRE ATT&CK® is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. The ATT&CK knowledge base is used as a foundation for the development of specific threat models and methodologies in the private sector, in government, and in the cybersecurity product and service community.

With the creation of ATT&CK, MITRE is fulfilling its mission to solve problems for a safer world — by bringing communities together to develop more effective cybersecurity. ATT&CK is open and available to any person or organization for use at no charge.

ATT&CK Matrix for Enterprise

layout: side | show sub-techniques | hide sub-techniques

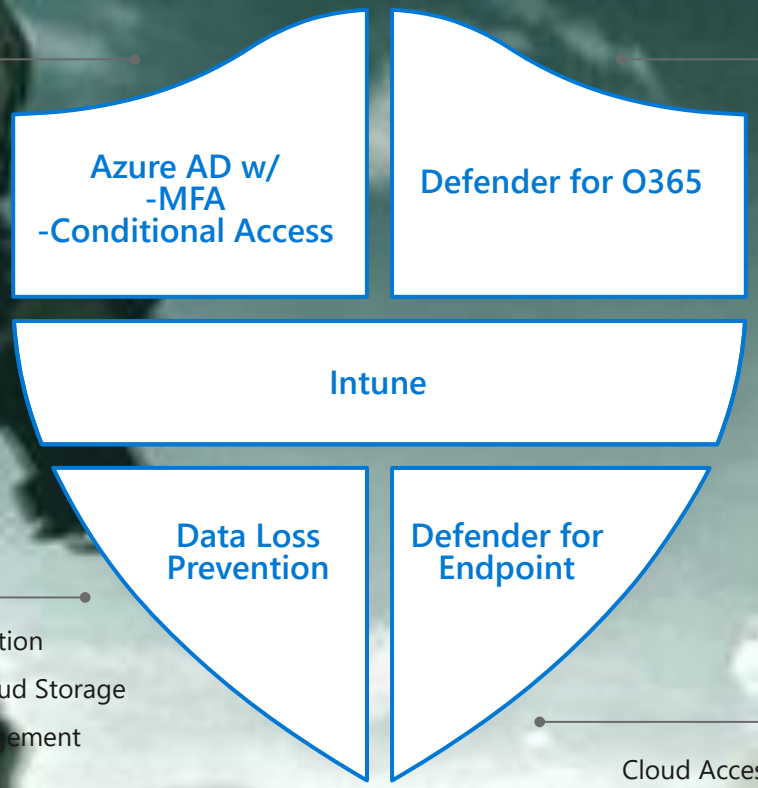
sance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	
ques	8 techniques	10 techniques	14 techniques	20 techniques	14 techniques	44 techniques	17 techniques	32 techniques	9 techniques	17 techniques	18 techniques	9 techniques	14
g (3)	Acquire Access	Content Injection	Cloud Administration Command	Account Manipulation (7)	Abuse Elevation Control Mechanism (6)	Abuse Elevation Control Mechanism (6)	Adversary-in-the-Middle (4)	Account Discovery (4)	Exploitation of Remote Services	Adversary-in-the-Middle (4)	Application Layer Protocol (5)	Automated Exfiltration (1)	Accou Remov
Host	Acquire Infrastructure (8)	Drive-by Compromise	Command and Scripting Interpreter (11)	BITS Jobs	Access Token Manipulation (5)	Access Token Manipulation (5)	Brute Force (4)	Application Window Discovery	Internal Spearphishing	Archive Collected Data (3)	Communication Through Removable Media	Data Transfer Size Limits	Data Destru
dentity	Compromise Accounts (3)	Exploit Public-Facing Application	Container Administration Command	Boot or Logon Autostart Execution (14)	Account Manipulation (7)	BITS Jobs	Credentials from Password Stores (6)	Browser Information Discovery	Lateral Tool Transfer	Audio Capture	Content Injection	Exfiltration Over Alternative Protocol (3)	Data E Impact
	Compromise Infrastructure (8)	External Remote Services	Deploy Container	Boot or Logon Initialization Scripts (5)	Boot or Logon Autostart Execution (14)	Build Image on Host	Exploitation for Credential Access	Cloud Infrastructure Discovery	Remote Service Session Hijacking (2)	Automated Collection	Data Encoding (2)	Exfiltration Over C2 Channel	Data Manip
Org	Develop Capabilities (4)	Hardware Additions	Exploitation for Client Execution	Browser Extensions	Boot or Logon Initialization Scripts (5)	Debugger Evasion	Forced Authentication	Cloud Service Dashboard	Remote Services (8)	Browser Session Hijacking	Data Obfuscation (3)	Exfiltration Over Other Network Medium (1)	Defac
	Establish Accounts (3)	Phishing (4)	Inter-Process Communication (3)	Compromise Host Software Binary	Create or Modify System Process (5)	Deobfuscate/Decode Files or Information	Forge Web Credentials (2)	Cloud Service Discovery	Replication Through Removable	Clipboard Data	Dynamic Resolution (3)	Exfiltration	Disk W
	Obtain Capabilities (7)	Replication Through	Native API	Create		Deploy Container	Input Capture (4)	Cloud Storage Object Discovery		Data from Cloud Storage			Endpo Servic
	Stage					Direct Volume Access		Container and					Financ
						Domain or Tenant Policy Modification (2)							



Consolidate security

Streamline and strengthen

Replace up to 40 disparate products with integrated, end-to-end security



- Single Sign on + Self-Service Reset
- Multifactor + Passwordless Authentication
- Conditional Access
- Privileged Access Management
- Identity Governance
- Active Directory

- Mobile Application Management
- Mobile Device Management

- Data Discovery
- Data Classification
- Data Loss Prevention
- Insider Risk Management
- Database Security
- Information and Messaging Encryption

- Device Encryption
- Encrypted Cloud Storage
- Secrets Management

- Endpoint Detection and Response
- Endpoint Protection Platform
- Forensic Tools
- Intrusion Prevention System
- Threat Vulnerability Management
- Anti-phishing
- User and Entity Behavior Analytics
- Threat Intelligence Feeds
- App and Browser Isolation
- Attachment Sandboxing

- Application Control
- End-user Training
- Network Firewall (URL Detonation)
- Host Firewall
- Secure Email Gateway
- Security Assessment
- SIEM
- SOAR
- Incident Response Services
- DDoS Protection
- IoT Protection

- Cloud Access Security Broker
- Cloud Workload Protection Platform
- Cloud Security Posture Management

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Microsoft Sentinel

Microsoft Sentinel | MITRE ATT&CK (Preview)

Selected workspace: 'sentinellaw'

Search

Favorites
- MITRE ATT&CK (Preview)
- Incidents
- Workbooks
- Data connectors
- Analytics
- Automation
- Logs

General

Threat management

Content management

Configuration
- Workspace manager (Preview)
- Data connectors
- Analytics
- Summary rules (Preview)
- Watchlist
- Automation
- Settings

Search by technique ID,... | Matrices type view : 13 selected | Coverage level : All | Active rules: 3 selected | Simulated rules: Analytics rule templates

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	Command And Control	Ex
10 techniques	8 techniques	9 techniques	14 techniques	19 techniques	13 techniques	42 techniques	17 techniques	31 techniques	9 techniques	17 techniques	16 techniques	9 t
2 +2 Active Scanning	Acquire Access	4 +4 Drive-by Compromise	12 +10 Command and Scripting...	30 +15 Account Manipulation	2 +2 Abuse Elevation Control...	Abuse Elevation Control...	1 +1 Man-in-the-Middle	13 +10 Account Discovery	4 +4 Exploitation of Remote Services	Man-in-the-Middle	6 +4 Application Layer Protocol	3
1 +1 Gather Victim Host...	Acquire Infrastructure	21 +15 Exploit Public-Facing...	6 +3 Exploitation for Client Execution	BITS Jobs	Access Token Manipulation	1 +1 Access Token Manipulation	38 +10 Brute Force	Application Window...	1 +1 Internal Spearphishing	Archive Collected Data	Communication Through...	6
2 +2 Gather Victim Identity...	Compromise Accounts	10 +10 External Remote Services	1 Inter-Process Communication	Boot or Logon Autostart...	Boot or Logon Autostart...	BITS Jobs	5 +4 Credentials from Password Stores	Browser Bookmark...	1 +1 Lateral Tool Transfer	Audio Capture	2 Data Encoding	2
3 +3 Gather Victim Network...	Compromise Infrastructure	Hardware Additions	Native API	Boot or Logon Initialization...	Boot or Logon Initialization...	Debugger Evasion	Exploitation for Credential...	Debugger Evasion	1 +1 Remote Service Session...	Automated Collection	1 Data Obfuscation	2
1 +1 Gather Victim Org Information	1 +1 Develop Capabilities	4 +4 Phishing	Scheduled Task/Job	1 +1 Browser Extensions	Create or Modify System...	3 +2 Deobfuscate/Decode Files or...	Forced Authentication	Device Driver Discovery	Remote Services	1 +1 Man in the Browser	9 +6 Dynamic Resolution	2
Phishing for Information	1 +1 Establish Accounts	Replication Through...	Shared Modules	2 +1 Compromise Client Softwar...	26 +24 Domain Policy Modification	Direct Volume Access	Forge Web Credentials	Domain Trust Discovery	Replication Through...	Clipboard Data	3 +2 Encrypted Channel	
Search Closed Sources	Obtain Capabilities	2 +2 Supply Chain Compromise	1 +1 Software Deployment...	22 +12 Create Account	Escape to Host	Domain Policy Modification	Input Capture	File and Directory...	1 +1 Software Deployment...	2 +1 Data from Information...	8 +7 Fallback Channels	8
1 +1 Search Open Technical...	Stage Capabilities	1 Trusted Relationship	System Services	1 +1 Create or Modify System...	Event Triggered Execution	1 +1 Execution Guardrails	3 +2 Modify Authentication...	Group Policy Discovery	Taint Shared Content	3 +3 Data from Local System	Ingress Tool Transfer	2

Drive-by Com

ID T1189

4 Detections

Description

Adversaries ma website over th the user's web adversaries ma exploitation be Token.Multiple including:A leg have injected s iFrames, and cr View full techn

Tactic

Initial Acce

Recommended

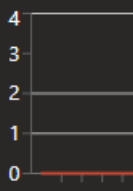
Active coverag

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Simulated cov

4 Analy

Incidents and



SecurityAlert (

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Features

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Collect

Detect

Investigate

Respond

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Features

Incidet Summary / Reporting

"A succinct report that outlines the context and environment, identifies open issues, and details protective measures, tailored to the tone and language suitable for the report's audience."

Reverse Engineering

"Translate and reverse engineer malware scripts into natural language to empower every team member to perform technical tasks."

Manage Security Policies

"Create a new policy, cross-reference it with existing ones to identify conflicts, and summarize current policies to efficiently manage complex organizational contexts."

Guided Investigations

"Acquire context for incidents to swiftly triage complex security alerts into actionable summaries and expedite remediation with step-by-step response guidance."

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Demo